Experiment #4

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Evolution of the Teluroncio Method: Comparative Analysis of Traction Configurations on the RENFE 313

Locomotive: RENFE 313 (ALCO DL-535-S) | **Test:** 20‰, 300 t | **Date:** June 2026

1. Introduction

The Teluroncio Method has evolved from its initial formulation as a purely empirical procedure (grade tests + iterative CK model adjustment) to incorporating theoretical calculation tools such as the Maxima v0.2 program. This experiment documents the transition between both phases, comparing the performance of the RENFE 313 locomotive under eight different configurations representing successive stages in the method's evolution.

2. Materials

Locomotive: RENFE 313 (ALCO DL-535-S), 83.9 t, 900 kW

Standardised test: CTN test route, 20‰ grade, 300 t trailing load

Conditions: Dry and rain

Wind: MaxWindSpeedMpS = 0 (controlled conditions)

Software: Open Rails NewYear MG 171.3

Tools: HUD (speed, tractive effort, power, throttle), logs, screenshots

3. Empirical Method

Eight configurations (A to H) were defined, combining three variables: origin of the tractive effort curve, CK model parameters, and weather condition.

3.1 Dry condition configurations

ID	Configuration	CK A	Tractive effort curve	Origin
A	Original MSTS	?	Original from .eng file	Uncalibrated file
B	Teluroncio 1	0.33	ALCO documentary	Validated calibration
C	Teluroncio 2a	0.33	Maxima μ_0=0.33	In-house program
D	Teluroncio 2b	0-33	Maxima μ_0=0.28	In-house program (documentary μ_0)

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3.2 Rain condition configurations

ID	Configuration	CK A	Tractive effort curve
E	Teluroncio 1	0.33	ALCO documentary
F	Teluroncio 2a	0.33	Maxima μ_0=0.33
G	Teluroncio 2b	0.33	Maxima μ_0=0.28
H	Teluroncio 2b coherent	0.28	Maxima μ_0=0.28

4. Experimental Results

4.1 Dry condition tests (20‰, 300 t)

ID	CK A	Curve	v (km/h)	TE (kN)	P (kW)	Throttle	Wheel slip?
A	?	MSTS	37.2	87.5	919	100%	No
B	0.33	Maxima 0.33	31.7	85.8	896	100%	No
C	0.33	Maxima 0.28	31.7	85.6	896	100%	No
D	0.33	Doc. ALCo	30.8	85.9	877	100%	No

4.2 Rain condition tests (20‰, 300 t, MaxWindSpeedMpS = 0)

ID	CK A	Curve	v (km/h)	TE (kN)	P (kW)	Throttle	Wheel slip?
E	0.33	Doc. ALCo	11.6	83.4	442	50%	Yes
F	0.33	Maxima 0.33	11.8	83.5	451	50%	Yes
G	0.33	Maxima 0.28	11.8	83.3	449	50%	Yes
H	0.28	Maxima 0.28	10.9	83.5	449	50%	Yes

4.3 Comparative Speed Overview — Dry vs. Rain (20‰, 300 t)

ID	Weather	Velocity (km/h)
A	🔅 Dry	37.2
B	🔅 Dry	30.8
C	🔅 Dry	31.7
D	🔅 Dry	31.7
E	🌧️ Rain	11.6
F	🌧️ Rain	11.8
G	🌧️ Rain	11.8
H	🌧️ Rain	10.9

5. Analysis

5.1 Dry condition comparison

Comparison	Difference	Interpretation
A vs B	+6.4 km/h (+21%)	The original MSTS file unrealistically inflates performance. Despite having the lowest tractive effort (189.3 kN vs. 246.9 kN), it achieves the highest speed, indicating artificially low running resistance and absence of a CK adhesion model.
C vs B	+0.9 km/h (+2.9%)	The Maxima curve (μ_0=0.33) offers slightly better performance than the ALCO documentary curve, within a reasonable margin.
C vs D	0.0 km/h	Under dry conditions, μ_0 does not influence equilibrium speed. Both configurations operate in the constant power zone ($P = F \times v$).

5.2 Rain condition comparison

Comparison	Difference	Interpretation
E vs F vs G	~0.2 km/h	With the same CK (A=0.33), the tractive effort curve does not influence rain performance. The CK model dominates over the curve.
G vs H	-0.9 km/h (-7.6%)	Lowering CK A from 0.33 to 0.28 causes a significant speed drop. Parameter A is the dominant factor under low-adhesion conditions.
E (new) vs E (old)	-3.3 km/h (-22%)	The old value (14.9 km/h) was likely obtained with wind enabled. Under controlled wind conditions (MaxWindSpeedMpS = 0), all configurations converge around 11–12 km/h.

5.3 Throttle behaviour

Under rain conditions, all configurations required reducing the throttle to 50% to prevent wheel slip. No significant differences were observed between the ALCO documentary curve and the Maxima curves in this regard.

5.4 Interaction between CK parameters

The experiment demonstrates that adhesion under adverse conditions does not depend exclusively on parameter C (climate modulator), but on the complete set (A, B, C). Reducing A (μ_0) from 0.33 to 0.28 causes a proportional decrease in available adhesion under all weather conditions, even with C held constant.

5.5 Safety margins in documentary curves

The ALCO documentary tractive effort curve includes safety margins inherent to the manufacturer's design criteria (cooling capacity, traction motor fatigue, operational limits). The Maxima program, being a purely physical model, does not incorporate these margins. This explains why, under rain conditions, the documentary curve (E) produces slightly more controlled behaviour than the Maxima curves (F, G), despite sharing the same CK parameters. The drawbar effort curve for the 20‰ grade, where the Davis resistance curve intersects the tractive effort curve at the locomotive's maximum speed, represents the practical operational limit that the documentary curve respects and the Maxima curve approaches without margin.

5.6 Wind effect

The discrepancy between the old value for test E (14.9 km/h) and the new one (11.6 km/h) is attributed to wind. Earlier tests were conducted with MaxWindSpeedMpS at its default value, potentially generating tailwind that reduced running resistance. The new tests were performed with MaxWindSpeedMpS = 0, ensuring controlled and reproducible conditions.

5.7 Incoherence between curve μ_0 and CK parameter A

Configuration G (CK A=0.33 + Maxima μ_0=0.28) represents a mismatch between the adhesion model and the tractive effort curve: the CK model expects a certain adhesion level (A=0.33), but the curve was generated with a lower coefficient (μ_0=0.28). Under dry conditions, this mismatch is irrelevant because the locomotive operates in the constant power zone. Under rain conditions, however, the CK model dominates and the curve becomes secondary: G performs identically to F (CK A=0.33 + Maxima μ_0=0.33). The practical consequence is that a curve generated with a given μ_0 must be paired with the matching CK parameter A. Otherwise, the curve provides no benefit and the CK model dictates behaviour alone.

6. Conclusions

#	Conclusion	Evidence
1	**The original MSTS file is unrealistic**	+21% speed with the lowest tractive effort. No CK adhesion model.
2	**Under dry conditions, Maxima curves match or improve the documentary curve**	+0.9 km/h (+2.9%) compared to B.
3	**Under dry conditions, μ_0 does not influence equilibrium speed**	C (0.33) = D (0.28).
4	**Under rain conditions, the CK model dominates over the curve**	E ≈ F ≈ G (same CK A=0.33), but G ≠ H (different CK A).
5	**CK-Curve coherence is essential for fine control**	H (coherent: CK 0.28 + Maxima 0.28) performs predictably.
6	**Parameter C does not act in isolation**	Rain adhesion depends on the complete set (A, B, C).
7	**Wind must be controlled to ensure reproducibility**	MaxWindSpeedMpS = 0 eliminates environmental variability.
8	**Documentary curves include safety margins absent from purely physical models**	Test E (documentary) shows slightly more controlled rain behaviour than F/G (Maxima).

7. Recommendation for the Teluroncio Method

"Under dry conditions, generating the curve with Maxima using the calibrated μ_0 (CK parameter A) offers the best performance. Under rain conditions, all configurations with CK A=0.33 perform equivalently (~11.7 km/h). Coherence between the curve μ_0 and CK parameter A is essential for predictable behaviour under adverse conditions. It is recommended to set MaxWindSpeedMpS = 0 for all standardised tests to ensure result reproducibility."

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8. References

- Open Rails Team. *Open Rails 1.6.1 Manual*, Release 1.6.1.0 (2026).

- Yuan, Z., Wu, M., Tian, C., Zhou, J., & Chen, C. (2021). *A Review on the Application of Friction Models in Wheel-Rail Adhesion Calculation*. Urban Rail Transit, 7(1), 1–11.

- Original technical documentation ALCO DL-535-S / RENFE 313 Series.

- Maxima v0.2 program (Teluroncio Method). Available at `/Curvas-Traccion/`.

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